

SMART INDIA HACKATHON 2026

PROJECT PROPOSAL REPORT

Hackathon:	Smart India Hackathon (SIH 2026)
Problem Statement ID:	SIH26017
Problem Statement Title:	Predictive Analytics System for Early Detection of Land Acquisition Delays
Organization / Ministry:	Ministry of Rural Development / Infrastructure & Smart Automation
Team Name:	Recursion
Team ID:	136984

EXECUTIVE SYNOPSIS: Land acquisition forms the foundational bottleneck for India's infrastructure expansion. Team Recursion proposes an AI-driven, explainable predictive decision-support platform designed to forecast structural bottlenecks and time overruns across the multi-stage land acquisition lifecycle. By synthesizing statutory mandates (RFCTLARR Act 2013 compliances, 3A/3D notifications, joint measurement surveys, disbursement metrics, and Rehabilitation & Resettlement friction) with spatial and legal datasets, the system identifies latency clusters before they crystallize into critical delays. Utilizing advanced Machine Learning (Survival Analysis and Gradient Boosted Trees) paired with Explainable AI (SHAP/LIME), this platform provides District Magistrates and Competent Authorities for Land Acquisition (CALA) with preemptive alerts, GIS-mapped high-

risk zones, and an automated mitigation playbook, ensuring capital efficiency and adherence to statutory timelines.

2. PROBLEM BACKGROUND & STATUTORY CONTEXT

The acquisition of land for national highways, railway freight corridors, and smart cities is governed primarily by the Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (RFCTLARR 2013) or the National Highways Act, 1956. The lifecycle is highly fragmented, requiring synchronization across Revenue Departments, Gram Sabhas, Judiciary, and Requisitioning Agencies. A failure at any single node cascades into multi-year delays, escalating project costs and stalling national development.

2.1 The Land Acquisition Lifecycle (RFCTLARR 2013)

The statutory process involves rigorous compliance gateways:

- **Section 4 to 9 (SIA):** Social Impact Assessment formulation and institutional review. Must establish the public purpose and minimize displacement.
- **Section 11 (Preliminary Notification):** Gazetted declaration of intent to acquire specific parcels. Freezes land transactions.
- **Section 15 (Hearing of Objections):** Statutory 60-day window for public grievances. Often overwhelmed by high litigation volumes.
- **Section 19 (Declaration):** Final declaration of acquisition, which must occur within 12 months of Section 11 to prevent statutory lapse of proceedings.
- **Section 21-23 (Award Inquiry):** Valuation, solatium calculation, and financial compensation award formulation.
- **Section 31 & 38 (Possession):** Disbursement of compensation followed by physical handover of land.

2.2 Root Cause Breakdown of Primary Delay Drivers

Through empirical analysis of past highway and railway projects, we have identified the following structural drivers that lead to project overruns averaging 3-4 years:

- **Title Deed & Cadastral Disputes:** Outdated *Khatauni* (Record of Rights), unmutated inheritance claims, and fragmented physical boundaries lead to disbursement freezes. Compensation cannot be released without clear title.
- **Judicial Injunctions:** High volumes of writ petitions challenging compensation multipliers, alignment choices, or R&R definitions result in High Court stay orders (injunctions), freezing entire corridors.
- **Inter-departmental Latency:** Lack of real-time API integrations between PFMS (Public Financial Management System), MoEFCC (Forest Clearance), and State Revenue portals creates massive administrative dead-time.
- **Rehabilitation & Resettlement (R&R) Friction:** Severe resistance from displaced Gram Sabhas during joint measurement surveys, often escalating into civil unrest.

3. PROPOSED SOLUTION & CORE INNOVATION

Team Recursion is engineering a preemptive, multi-variate intelligence layer—an **Early-Warning System (EWS)**—that transitions the Ministry from reactive firefighting to proactive bottleneck mitigation. This platform serves as a digital twin for the entire acquisition lifecycle.

3.1 Core Value Proposition & Differentiators

- **Continuous Multi-Variate Risk Scoring:** Instead of static reporting, the system constantly computes a time-decaying probability of default for every statutory milestone. Models re-evaluate risks daily as new data (e.g., a court filing) enters the system.
- **Explainable AI (XAI) for Administrators:** District Collectors do not need a black-box percentage. Using TreeSHAP, the system translates a "78% delay risk" into specific textual drivers (e.g., "Risk driven by: 3 active civil suits + 40% unmutated land records"). This builds administrative trust.
- **Automated Mitigation Playbook:** The platform cross-references the predicted delay driver against the RFCTLARR statutory playbook, recommending precise administrative actions (e.g., "Convene Section 64 Alternate Dispute Authority").
- **Geospatial (GIS) Integration:** Spatial visualization of acquisition alignments overlaid with "delay heatmaps," highlighting high-risk zones at the Tehsil/Village level to optimize resource allocation.

4. SYSTEM ARCHITECTURE & DATA PIPELINE

The solution utilizes a decoupled, Microservices-based architecture designed for high availability, fault tolerance, and horizontal scalability across State Data Centers.

4.1 Data Ingestion Sources

The system acts as an aggregation nexus for highly disparate state and central datasets:

- **Land Records (Bhulekh/AnyRoR):** Polled via API/Webhooks for ownership structures and mutation status.
- **eCourts APIs:** Scraped/integrated for identifying active writ petitions and stay orders on specific survey numbers.
- **PFMS & Treasury:** Tracking the velocity of compensation disbursement to identify escrow bottlenecks.
- **Bhuvan / Spatial DBs:** Cadastral GeoJSON layers and forest boundary intersections.

4.2 High-Level System Architecture Flow

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[LEVEL 0: CONTEXT]
STATE PORTALS (Bhulekh, eCourts, PFMS)  ---> [ LandGuard AI Core ] ---> DISTRICT COLLECTOR
/ MO-RD
(Raw Ownership, Court Stays, Finance)                                     (Risk Alerts,
Heatmaps, Playbook)

[LEVEL 1: COMPONENT ARCHITECTURE]
+-----+ +-----+ +-----+
---+
| 1. DATA INGESTION LAYER | | 2. FEATURE PIPELINE | | 3. ML INFERENCE ENGINE |
| | | | |
| - REST API Gateways | --> | - Pandas / Spark ETL | --> | - XGBoost Classifiers |
| | | | |
| - Celery Async Tasks | | - Spatial Joins (PostGIS) | | - Cox Survival Models |
| | | | |
| - Apache Kafka Streams | | - Missing Value Imputer | | - SHAP Explainer |
| | | | |
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| | |
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| 5. PRESENTATION TIER | | 4. MITIGATION ENGINE | <-----+
| - Next.js Web Dashboard | <-- | - Rules-based Expert Sys |
| - Leaflet.js GIS Engine | | - SLA Tracker |
+-----+ +-----+ +-----+
```

5. AI/ML METHODOLOGY & ALGORITHMIC FORMULATIONS

Predicting statutory delays is fundamentally a time-to-event problem layered with categorical risks. We propose a **Multi-Task Hybrid Model** architecture that models both absolute probability of breach and estimated time overrun.

5.1 Model Strategy

- **Survival Analysis (Cox Proportional Hazards / Random Survival Forests):** Used to estimate the expected time-to-completion for continuous milestones (e.g., the gap between Section 11 and Section 19). Survival models excel at handling right-censored data (projects currently ongoing).
- **Gradient Boosted Classifiers (XGBoost / LightGBM):** Used for binary breach predictions—calculating the explicit probability that a project will breach its statutory lapse date. Ensembles are robust to non-linear relationships in tabular data.
- **Graph Neural Networks (GNNs):** (Future Scope) To model complex dependency graphs between intersecting administrative departments (e.g., Forestry dependencies blocking Revenue progression).

5.2 Mathematical Formulations

The composite project risk score $(R_{\{proj\}})$ is calculated as a weighted sum of milestone breach probabilities, penalized by judicial and administrative friction. It dynamically adjusts as the project progresses:

$$R_{\{proj\}} = \sum (w_i * P(M_i_breach | X)) + \lambda * (C_{\{injunct\}} + C_{\{unmutated\}})$$

Where (w_i) represents the critical path weight of milestone (i) , (X) represents the feature vector of the project, and (λ) is the judicial penalty scalar.

The time-to-event is modeled via the hazard rate $(h(t|x))$, representing the instantaneous risk of delay given project covariates (x) :

$$h(t | x) = h_0(t) * \exp(\sum \beta_j x_j)$$

Where $(h_0(t))$ is the baseline hazard function, and (β_j) are the coefficients associated with the covariates (x_j) .

5.3 Explainability via SHAP

To establish trust with domain officers, the model outputs localized feature attribution using **TreeSHAP** (SHapley Additive exPlanations). Instead of a black box, the model mathematically distributes the risk prediction among the input variables.

$$\varphi_i(\mathbf{f}, \mathbf{x}) = \sum [(|S|! * (M - |S| - 1)!) / M!] * [\mathbf{f}_{\mathbf{x}}(S \cup \{i\}) - \mathbf{f}_{\mathbf{x}}(S)]$$

Where φ_i is the Shapley value for feature i , S is a subset of features not containing i , M is the total number of features, and $\mathbf{f}_{\mathbf{x}}(S)$ represents the expected model output given the features in S .

6. GIS & SPATIAL RISK MAPPING

A tabular interface is insufficient for linear infrastructure projects (e.g., 500km highways). The platform ingests GeoJSON/Shapefiles into **PostGIS**, overlaying cadastral (land parcel) boundaries with external risk vectors to create dynamic spatial intelligence.

- **Environmental Intersections:** Proximity to eco-sensitive zones, national parks, or reserved forests causing mandatory MoEFCC clearance delays. Buffer zones are calculated dynamically.
- **Displacement Density:** High concentrations of affected structures/villages plotted as heatmaps to predict R&R resistance and potential protest flashpoints.
- **Jurisdictional Boundaries:** Identifying administrative bottlenecks where the highway alignment crosses multiple poorly-coordinated Tehsils or Districts.

The dashboard utilizes *Leaflet.js / Mapbox GL* to render these spatial insights as dynamic Choropleth maps highlighting red-zone villages, allowing administrators to direct resources precisely where spatial friction is highest.

7. INTERACTIVE DECISION SUPPORT DASHBOARD & UI/UX

The user interface is designed for rapid comprehension by non-technical bureaucrats. It serves different echelons of the administration via strictly enforced **Role-Based Access Control (RBAC)**.

USER ROLE	ACCESS & CAPABILITIES
Central Ministry / MoRD	State-wide aggregate KPIs, macro-level capital blockage analytics, and multi-state infrastructure velocity trends. Focus on policy adjustments.
District Magistrate (DM/DC)	District-level breakdown, active SHAP delay drivers, pending file counts by Tehsil, and authority to trigger high-level interventions.
Land Acquisition Collector (CALA)	Granular village-level tracking, litigation alerts per survey number, automated notices generation.
Project Proponent (NHAI/Railways)	Read-only spatial tracking, disbursement status, and AI-predicted physical possession handover timelines.

Key Dashboard Modules:

- **Executive KPI Cards:** Real-time metrics on funds disbursed vs. funds escrowed, active litigations, and overall project health scores.

- **Milestone Delays Gantt with AI Projections:** Compares baseline statutory timelines against AI-predicted delays, visualizing critical path breaches before they happen.
- **Litigation Tracker:** Auto-flags survey numbers with active High Court stays and tracks next hearing dates.
- **Automated Red-Flag Notifications:** Automated SMS/Email dispatches 30, 60, and 90 days prior to statutory lapse dates (e.g., Section 19 deadlines).

8. TECH STACK SPECIFICATION

The architecture is strictly composed of enterprise-grade, open-source technologies to ensure high performance, security, and zero vendor lock-in for the Government of India.

- **Frontend (Client-Side):** Next.js (React 18) for server-side rendering, TailwindCSS for responsive design, Recharts for analytics, Leaflet.js / Mapbox GL for WebGIS.
- **Backend (API & Logic):** FastAPI (Python 3.11) for high-throughput ML inference and API routing, Node.js for real-time WebSockets and notification pushing.
- **Database & Storage:** PostgreSQL 16 with PostGIS extension for relational & spatial data, Redis for caching JWTs & heavy analytical queries, MinIO (S3-compatible object storage) for document storage (Notifications/Gazettes).
- **Machine Learning / MLOps:** XGBoost, Scikit-Learn, Lifelines (Survival Analysis), SHAP, MLflow for model registry and drift monitoring.
- **DevOps & Deployment:** Docker for containerization, Kubernetes (K8s) for orchestration, Celery + RabbitMQ for background asynchronous workers.

9. RISK MITIGATION PLAYBOOK & INTERVENTIONS

The system does not just identify problems; it maps predictive drivers directly to actionable, statutory remediations, effectively serving as an expert legal assistant for the CALA.

IDENTIFIED BOTTLENECK (SHAP OUTPUT)	PRESCRIPTIVE AUTOMATION DIRECTIVE	TARGET SLA
Compensation Disbursal < 40% (Title Disputes)	Auto-schedule Tehsil-level Revenue Camp for mutation clearance. Issue automated notices to conflicting heirs under Section 64.	14 Days
Active Judicial Injunction (Stay Order)	Trigger priority alert to State Standing Counsel. Auto-generate docket summary for rapid vacating petition filing.	7 Days
Section 11 to 19 Gap > 300 Days	CRITICAL ALERT: Impending statutory lapse. Escalate directly to District Magistrate dashboard for emergency review.	3 Days
High R&R Friction (Large affected population)	Initiate early tripartite dialogue. Propose alternative resettlement packaging to Gram Sabha preemptively.	21 Days

10. IMPLEMENTATION ROADMAP, FEASIBILITY & DB SCHEMA

The proposed system is highly feasible given the current digitization thrust of the Digital India Land Records Modernization Programme (DILRMP).

4-Phase Implementation Rollout Plan

- **Phase 1 (Months 0-3):** Data synthesis and schema design. Pilot deployment in 2 high-volume districts. Integration with static CSV extracts.
- **Phase 2 (Months 4-6):** State-wide expansion. Training the ML models on state-specific historical timelines. State-wide UI rollout.
- **Phase 3 (Months 7-9):** Live API synchronization with state Bhulekh portals and eCourts.
- **Phase 4 (Months 10-12):** Multi-state federation and handover to NIC (National Informatics Centre) for sovereign cloud hosting.

Security & Compliance: All Personal Identifiable Information (PII) regarding landowners will be masked at the application layer. Data will be encrypted in transit (TLS 1.3) and at rest (AES-256), strictly adhering to the Digital Personal Data Protection (DPDP) Act 2023.

11. TEAM RECURSION ROLE MATRIX

Our team comprises multi-disciplinary expertise tailored for full-stack AI deployment. Team ID: 136984.

ROLE DESIGNATOR	CORE RESPONSIBILITIES
Team Lead / Data Scientist	Architecture design, XGBoost modeling, SHAP integration, and mathematical formulation.
Backend & Cloud Architect	FastAPI microservices, PostgreSQL schema optimization, Dockerization, CI/CD pipeline.
Frontend UI/UX Developer	Next.js dashboard engineering, Tailwind responsive design, data visualization.
GIS & Spatial Data Engineer	PostGIS queries, Leaflet.js map rendering, shapefile polygon intersection logic.
Domain / Legal Specialist	RFCTLARR 2013 statutory rule mapping, prescriptive playbook generation, edge-case testing.

12. CONCLUSION & FUTURE SCOPE

Team Recursion's Predictive Analytics System translates opaque, bureaucratic inertia into quantifiable, mathematically derived risk metrics. By providing a scalable, explainable Early Warning System, the platform empowers the Ministry of Rural Development to safeguard thousands of crores in public infrastructure capital from statutory lapses and litigation lockups.

Future Scope: Expanding the architecture via Federated Learning to train globally robust models across disparate state land registries without moving sovereign data. Furthermore, integrating this engine as a microservice directly into the *PM GatiShakti National Master Plan* would provide unparalleled, predictive intelligence for India's trillion-dollar infrastructure pipeline, enabling true smart infrastructure orchestration.